

IDAMAN PHARMA MANUFACTURING

IDAMAN PHARMA PENICILLIN V TABLET 125MG

DESCRIPTION

White, clean, dry, round, uniform, uncoated tablet, biconvex tablet, scored on one side.

COMPOSITION

Each tablet contains Phenoxymethylpenicillin Potassium equivalent to 125 mg of Phenoxymethylpenicillin.

PHARMACODYNAMIC

Phenoxymethylpenicillin is a broad spectrum beta-lactam antibiotic with bactericidal action against Gram-positive bacteria and Gram-negative cocci. Its antimicrobial action is similar to that of benzylpenicillin. Phenoxymethylpenicillin is usually active against the following organisms:

a) Gram-positive aerobes and anaerobes including: *Bacillus anthracis*, *Clostridium perfringens*, *Clostridium tetani*, *Corynebacterium diphtheriae*, *Erysipelothrix rhusiopathiae*, *Listeria monocytogenes*, *Peptostreptococcus spp.*, *Streptococcus agalactiae* (Group B), *Streptococcus pneumoniae*, and *Streptococcus pyogenes* (Group A).

b) Gram-negative including *Neisseria meningitidis* and *Neisseria gonorrhoeae*

Phenoxymethylpenicillin is inactivated by penicillinase and other beta-lactamases. Phenoxymethylpenicillin binds to penicillin binding proteins located on the inner membrane of the bacterial cell wall. Phenoxymethylpenicillin binds to and inactivates these proteins resulting in weakening of the bacterial cell wall and lysis.

PHARMACOKINETICS

Absorption: Phenoxymethylpenicillin is stable under acidic conditions so it can be administered by oral route. Phenoxymethylpenicillin is rapidly, but incompletely absorbed after oral administration and the absorption level is around 60%. The simultaneous administration of food slightly decreases the peak plasma concentration of phenoxymethylpenicillin, but does not appear to affect the extent of absorption. Peak plasma concentrations are reached in about

45 minutes. The peak plasma concentration increases approximately in proportion with increased doses. Peak serum concentrations of 3-6 jig per ml have been seen following dosage of 250 mg to 500 mg by mouth.

Distribution: Phenoxymethylpenicillin is widely distributed in the body tissues and fluids (volume of distribution about 0.2 l kg⁻¹ of body weight) and more readily penetrates inflamed tissues. It also diffuses across the placenta into foetal circulation and small amounts appear in the milk of nursing mothers. Eighty per cent is reported to be protein bound.

Biotransformation: Phenoxymethylpenicillin is partially metabolised to inactive penicilloic acid by hydrolysis of the lactam ring. This metabolism occurs in the liver.

Elimination: The plasma half-life of phenoxymethylpenicillin is about 45 minutes which may increase to four hours in renal failure. Excretion is by tubular secretion into urine. About 40% of the dose is eliminated in the urine either as under unchanged or as penicilloic acid in the first 10 hours after oral administration. Small excretion occurs in bile. Impaired absorption is seen in patients with coeliac disease.

INDICATIONS

Disease which are caused by penicillin-susceptible organisms, such as angina tonsillaris, bacterial bronchitis, acute otitis media, acute sinusitis, scarlet fever, furuncles, abscesses; supportive therapy in bacterial pneumonia in as much as parenteral therapy is not necessary; infection or danger of infection by haemolytic streptococci; for the prophylaxis of recurrences in rheumatic fever, polyarthritis, bacterial endocarditis, glomerulonephritis, particularly after tonsillectomy; prophylaxis against infection in stomatology (e.g. tooth extractions).

CONTRAINDICATIONS

Phenoxymethylpenicillin is contraindicated in patients with known penicillin hypersensitivity. Attention should be paid to possible cross-sensitivity with other beta-lactam antibiotics. Severe acute infections should not be treated with phenoxymethylpenicillin.

ADVERSE EFFECTS

Hypersensitivity

Potential allergic reactions include skin rashes, maculopapular rashes, angioedema, urticaria, angioneurotic oedema, erythema

multiforme, exfoliative dermatitis, fever, joint pain, serum sickness-like reactions, haemolytic anaemia, interstitial nephritis or anaphylactic shock (which could be fatal) with collapse and anaphylactoid reactions (asthma, purpura, gastrointestinal symptoms). Although these are less common, and take a milder course, in oral treatment than during parenteral penicillin treatment, it should be remembered that all degrees of hypersensitivity, including fatal anaphylaxis, have been observed with oral penicillin.

Gastrointestinal tract

Phenoxymethylpenicillin potassium is generally well tolerated. Occasionally soft stools occur and they do not require the interruption of the treatment.

Nausea, diarrhoea, vomiting, stomatitis and glossitis are sometimes seen. Sustained severe diarrhoea should prompt suspicion of pseudomembranous colitis. As this condition may be life threatening, phenoxymethylpenicillin should be withdrawn immediately and treatment guided by bacteriologic studies with appropriate antibiotherapy. Sore mouth or tongue and black hairy tongue are other gastrointestinal effect.

Blood

Eosinophilia, haemolytic anaemia, leukopenia, thrombocytopenia and agranulocytosis are extremely rare. Other possible effects on the blood composition include: neutropenia, haemolytic anaemia and coagulation disorders.

Central nervous system

Central nervous system toxicity, including convulsions, has been reported, especially following high doses or in severe renal impairment. Paraesthesia has been reported with prolonged use.

Prolonged use

As with other broad-spectrum antibiotics prolonged use may result in the overgrowth of non-susceptible organisms, e.g. candida. This may present a vulvo-vaginitis.

Electrolyte disturbances

Electrolyte disturbances because of the large amounts of potassium given when phenoxymethylpenicillin potassium is used. Phenoxymethylpenicillin potassium in high doses may lead to hyperkalaemia, particularly in those with renal impairment.

In addition, many semisynthetic penicillins (including carbenicillin, cloxacillin, mezlocillin, nafcillin, piperacillin, and ticarcillin) can act as non-absorbable anions in the distal tubule, resulting in urinary potassium loss and hypokalaemia. This effect may be mediated by volume depletion, and has been mainly in severely ill patients.

Jarisch-Herxheimer reaction

Some patients with syphilis and other spirochaete infections may have a Jarisch-Herxheimer reaction shortly after starting treatment with penicillin, which is probably due to the release of endotoxins from the killed treponemes and should not be mistaken for a hypersensitivity reaction. Symptoms include fever, chills, headache, and reaction at the site of the lesions. The reaction can be dangerous in cardiovascular syphilis, or where there is a serious risk of increased local damage, such as with optic atrophy.

Others

Hepatitis and cholestatic jaundice have been reported rarely with many penicillins, notably penicillinase-resistant penicillins such as flucloxacillin and oxacillin, and also combinations of amoxicillin or ticarcillin with clavulanic acid. Neuropathy and nephropathy have also been reported.

WARNING AND PRECAUTION

1) Phenoxymethylpenicillin should be given with caution to patients with a history of allergy, especially to other drugs. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Penicillin V Tablet, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Penicillin V Tablet must be discontinued immediately and appropriate alternative therapy instituted.

2) Particular caution should be exercised in prescribing phenoxymethylpenicillin to patients with an allergic diathesis or with bronchial asthma.

3) Oral penicillins are not indicated in patients with severe illness or with a gastrointestinal disease that causes persistent nausea, vomiting, gastric dilation, cardiospasm, intestinal hypermotility or diarrhoea because absorption may be reduced. Occasionally, patients do not absorb therapeutic amounts of orally administered penicillin.

4) Streptococcal infections should be treated for a minimum of 10 days and post-therapy cultures should be performed to confirm the eradication of the organisms.

5) In patients undergoing long-term phenoxymethylpenicillin treatment the complete and differential blood count, as well as the liver and kidney function, should be monitored. During long-term treatment attention should also be paid to the potential overgrowth of resistant organisms including *Pseudomonas* or *Candida*. If super-infection occurs, appropriate measures should be taken.

6) Caution should be used when treating patients with a history of antibiotic-associated colitis.

7) High doses should be used with caution in patients receiving potassium-containing drugs or potassium sparing-diuretics.

8) Care is necessary if very high doses of penicillins are given, especially if renal function is poor, because of the risk of neurotoxicity. The intrathecal route should be avoided. Renal, hepatic and haematological status should be monitored during prolonged and high-dose therapy. In renal impairment the safe dosage may be lower than usually recommended.

9) Under certain circumstances penicillin may interfere with some diagnostic tests such as direct antiglobulin (Coombs) tests, tests for serum proteins and non-enzymatic glucose tests using copper sulfate may be false-positive, whereby can stimulate glycosuria. Penicillins may also interfere with tests that use bacteria, for example the Guthrie test for phenylketonuria using *Bacillus subtilis* organisms.

10) Not recommended for chronic, severe or deep-seated infections eg, subacute bacterial endocarditis, meningitis or syphilis.

11) Should be used in cautions with patient having seizure disorders.

12) Desensitisation may be attempted if treatment with a penicillin is considered essential.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

None known.

USAGE DURING PREGNANCY AND LACTATION

Pregnancy: Risk-benefit must be considered when phenoxymethylpenicillin is intended for use in pregnancy.

Lactation: Breast feeding is not contraindicated with phenoxymethylpenicillin potassium. Trace quantities of phenoxymethylpenicillin potassium can be detected in breast milk. While adverse effects are

apparently rare, two potential problems exist for nursing infant are modification of bowel flora and direct effects on the infant such as allergy/sensitisation. Caution should therefore be exercised when prescribing for the nursing mother.

DRUG INTERACTIONS

1) As penicillins like phenoxymethylpenicillin are only active against proliferating microorganisms, phenoxymethylpenicillin should not be combined with bacteriostatic antibiotics such as tetracycline, erythromycin, chloramphenicol and sulphonamides.

2) Concomitant use of uricosuric drugs (e.g. probenecid and sulfinpyrazone) reduces the excretion of phenoxymethylpenicillin resulting in increased plasma levels and thus prolongs its action.

3) Phenoxymethylpenicillin may reduce the excretion of methotrexate causing an increased risk of toxicity.

4) Guar gum may slow the speed of absorption of phenoxymethylpenicillin.

5) Absorption of phenoxymethylpenicillin reduced by neomycin.

6) Combined use of phenoxymethylpenicillin and oral anticoagulants (e.g. warfarin) may prolong prothrombin time.

7) Common experience in anticoagulant clinics is that INR can be altered by a course of broad-spectrum penicillins such as ampicillin, although studies have failed to demonstrate an interaction with coumarins and phenindione.

8) Phenoxymethylpenicillin may inactivate oral typhoid vaccine if ingested concomitantly.

9) May reduce the efficacy of oral contraceptives.

10) Reduced excretion with probenecid and sulfinpyrazone.

11) Potentially Fatal: Increased risk of anaphylactic reactions with nadolol and propranolol.

12) Absorption may be slightly affected by food.

13) False-positive urine glucose reactions with Clinitest, Benedict solution, or Fehling solution. It may also caused positive Coombs test.

14) Clinically significant reductions in renal clearance have also been noted for amoxicillin, nafcillin and ticarcillin.

RECOMMENDED DOSAGE

Usual Adult Dose : Oral, the equivalent of phenoxymethylpenicillin - 125 to 500 mg (200,000 to 800,000 units) every six to eight hours.

Note : Continuous prophylaxis of streptococcal infections in patients with a history of rheumatic heart disease and/or chorea - Oral, the equivalent of phenoxymethylpenicillin - 125 to 250mg (200,000 to 400,000 unit) every twelve hours.

Usual Adult Prescribing Limits : The equivalent of phenoxymethylpenicillin - up to 7.2 gm.

Usual Paediatric Dose : Oral, the equivalent of phenoxymethylpenicillin infant and children up to 12 years of age : 2.5 to 9.3 mg per kg body weight every 4 hours; 3.75 to 14 mg per kg body weight every 6 hours; 5.0 to 18.7 mg per kg body weight every 8 hours.

Each tablet should be swallowed whole with water, at least 30 minutes before food or 2 hours after food, as ingestion of phenoxymethylpenicillin with meals slightly reduces the absorption of drug.

ROUTE OF ADMINISTRATION : Oral
SYMPTOMS AND TREATMENT FOR OVERDOSAGE

Symptoms: Severe nausea, vomiting, diarrhoea, stomach pain, confusion, drowsiness and myoclonus which may progress to convulsion. Large dose may also cause hyperkalemia, particularly for patients with renal insufficiency. Rarely major motor seizures may occur. If other symptoms are present, consider the possibility of an allergic reaction.

Treatment: There is no known antidote. Symptomatic and supportive therapy is recommended. May administer activated charcoal with a cathartic, for example sorbitol to hasten drug elimination. It is advisable to monitor blood levels in patients with renal malfunction. Haemodialysis may aid in removing penicillins from the blood.

PRESENTATION
Idaman Pharma Penicillin V Tablet 125mg
(Malaysia Reg. No: MAL19860070AZ)

Packed in blister of 10 tablets in carton box (50 x 10's)

STORAGE CONDITIONS

Keep container tightly closed in a dry place, below 30°C; Protect from light.

SHELF-LIFE : 2 years from the date of manufacture.

For further information, please consult your doctor or your pharmacist.

Date of Revision : 14 Dec 2019

PRODUCT REGISTRATION HOLDER / MANUFACTURER:
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